TEST REPORT

Application No.:	KSEM2306001405MD
Applicant:	Richbeam (Suzhou) Technology Co.,LTD
Address of Applicant:	NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Manufacturer:	Richbeam (Suzhou) Technology Co.,LTD
Address of Manufacturer:	NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Factory:	Richbeam (Suzhou) Technology Co.,LTD
Address of Factory:	NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Equipment Under Test (EUT):	
EUT Name:	2D TOF LIDAR
Model No.:	LoraBeam1H, LoraBeam1, LoraBeam1S, KoraBeam1, KoraBeam1S ♣
♣	Please refer to section 2 of this report which indicates which model was actually tested and which were electrically identical.
Standard(s) :	EN IEC 61496-1: 2020 Clause 4.3.2 & 5.4.3
Date of Receipt:	2023-06-21
Date of Test:	2023-07-07 to 2023-08-18
Date of Issue:	2023-08-21
Test Result:	**Pass***

* In the configuration tested, the EUT complied with the standards specified above.

Eric Lin
Laboratory Manager

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Revision Record			
Version	**Description**	***Date***	***Remark***
00	Original	2023-08-21	/

Authorized for issue by:				
		Terry Hou/Project Engineer		
		Eric Lin /Reviewer		

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2 Test Summary

Immunity Part				
Item	**Standard**	**Method**	**Requirement**	**Result**
Conducted Immunity at DC Power Port (150kHz-80MHz)	EN IEC 61496-1: 2020	EN 61000-4-6:2014	10Vrms (emf),80%,1kHz Amp. Mod.	Pass
Electrical Fast Transients Burst at DC Power Port		EN 61000-4-4:2012	1kV 5/50ns Tr/Td 5kHz or 100kHz Repetition Frequency	Pass
Electrostatic Discharge		EN 61000-4-2:2009	±4kV Contact Discharge, ±8kV Air Discharge	Pass
Radiated Immunity (80MHz-1GHz, 1.4GHz-6GHz)		EN IEC 61000-4-3: 2020	80MHz to 1GHz 10V/m, 80%, 1kHz Amp. Mod. 1.4 to 6GHz 3V/m, 80%, 1kHz Amp. Mod.	Pass
Surge at DC Power Port		EN 61000-4-5:2014+A1:2017	1.2/50μs Tr/Td , 0.5kV Line to Line , 1kV Line to Ground	Pass
Voltage Dips and Interruptions		EN 61000-4-29:2000	0 % UT for 0.5per;0 % UT for 1per;0 % UT for 250per;70 % UT for 25per;UT is Supply Voltage	Pass

Declaration of EUT Family Grouping:

There are series models mentioned in this report and they are the Identical in electrical and electronic characters. Only the model LoraBeam1 was tested since their differences were the model number and appearance.

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4 General Information

4.1 Details of E.U.T.

	Power supply:	DC 9V-36V Test Voltage: DC 12V
	Cable(s):	Cable 1m

4.2 Description of Support Units

Description	Manufacturer	Model No.	Serial No.
--	--	--	--
The EUT has been tested as an independent unit.			

4.3 Measurement Uncertainty & Decision Rule

Measurement Uncertainty:

No.	Item	Measurement Uncertainty (U_{LAB}) *	U_{CISPR}
1	Conducted Emission at mains port using AMN	2.4dB (9kHz to 150kHz)	3.8dB (9kHz to 150kHz)
		2.2dB (150kHz to 30MHz)	3.4dB (150kHz to 30MHz)
2	Conducted Emission at telecommunication port using AAN	4.0 dB (150kHz to 30MHz)	5.0dB (150kHz to 30MHz)
3	Radiated Power	3.2dB	4.5dB (30MHz to 300MHz)
4	Radiated Emission (10m)	4.1 dB	6.3dB (30MHz-1GHz)
5	Radiated Emission (3m)	4.6 dB (30MHz-1GHz)	6.3dB (30MHz-1GHz)
		5.0dB (1GHz-6GHz)	5.2dB (1GHz-6GHz)
		5.2dB (6GHz-18GHz)	5.5dB (6GHz-18GHz)
		5.3dB (18GHz-40GHz)	N/A

Note: The measurement uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

Decision Rule:

• CISPR 16-4-2 for emission measurements is as below described.

Pass means the test result passed the test standard requirement, please find the detailed decision rule in the report relative section.

U_{LAB} less than U_{CISPR}, therefore:

– compliance is deemed to occur if no measured disturbance level exceeds the disturbance limit.

– non-compliance is deemed to occur if any measured disturbance level exceeds the disturbance limit.

• For immunity testing no decision rule is applicable.

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4.4 Test Location

All tests were performed at:

Compliance Certification Services (Kunshan) Inc.

No.10 Weiye Rd, Innovation park, Eco&Tec, Development Zone, Kunshan City, Jiangsu, China.

Tel: +86 512 5735 5888 Fax: +86 512 5737 0818

No tests were sub-contracted.

Note:

1.SGS is not responsible for wrong test results due to incorrect information (e.g., max. internal working frequency, antenna gain, cable loss, etc) is provided by the applicant. (If applicable).

2.SGS is not responsible for the authenticity, integrity and the validity of the conclusion based on results of the data provided by applicant. (If applicable).

4.5 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

• A2LA

Compliance Certification Services (Kunshan) Inc. is accredited by the American Association for Laboratory Accreditation (A2LA). Certificate No. 2541.01.

• ISED

Compliance Certification Services (Kunshan) Inc. has been recognized by Innovation, Science and Economic Development Canada (ISED) as an accredited testing laboratory. Company Number: 2324E

• VCCI

The 3m and 10m Semi-anechoic chamber and Shielded Room of Compliance Certification Services (Kunshan) Inc. has been registered in accordance with the Regulations for Voluntary Control Measures with Registration No.: R-20134, R-11600, C-11707, T-11499, G-10216 respectively.

4.6 Deviation from Standards

None

4.7 Abnormalities from Standard Conditions

None

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5 Equipment List

Conducted Immunity at DC Power Port (150kHz-80MHz)					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
Test System for Conducted and Radiated Immunity	TESEQ	NSG 4070B	KSZ201705E003	02/03/2023	02/02/2024
Amplifier	TESEQ	SCCXE75	KSZ201705E004	02/03/2023	02/02/2024
EM-Koppelzange	SCHAFFNER	KEMZ 801	CZ301002	02/03/2023	02/02/2024
Attenuator	EURO MC	7860 ORGEVAL	CZ301084	03/16/2023	03/15/2024
Directional Coupler	HIGH POWER	C21A8	CZ750021	09/06/2022	09/05/2023
CDN（Coupling and Decoupling Network)	SCHAFFNER	CDN M216	CZ301085	03/16/2023	03/15/2024
CDN（Coupling and Decoupling Network)	SCHAFFNER	CDN M316	CZ301025	03/16/2023	03/15/2024
CDN（Coupling and Decoupling Network)	TESEQ	CDN S751	KS301184-2	03/16/2023	03/15/2024
CDN（Coupling and Decoupling Network)	TESEQ	CDN M116	KS301184-1	03/16/2023	03/15/2024
CDN	TESEQ	CDN T2-10S	KS301286	03/16/2023	03/15/2024
CDN	TESEQ	CDN T4-10S	KS301287	03/16/2023	03/15/2024
CDN	TESEQ	CDN T8-10S	KS301288	03/16/2023	03/15/2024
Current Clamp	TESEQ	IP-DR250	KS201703E001	02/03/2023	02/02/2024
CDN	TESEQ	CDN M432	KUS2003M001-1	02/03/2023	02/02/2024
CDN	TESEQ	CDN M432-3LN	KUS2003M001-2	02/03/2023	02/02/2024
CDN	TESEQ	CDN M532	KUS2003M001-3	02/03/2023	02/02/2024
CDN	TESEQ	CDN M232	KSZ201706E001	03/16/2023	03/15/2024
CDN	TESEQ	CDN M332	KSZ201706E002	03/16/2023	03/15/2024
Software	TESEQ	NSG 4070-v 1.3.0.1	N/A	N/A	N/A

Electrical Fast Transients Burst at DC Power Port					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	Transient2000	KS301188-1	08/22/2022	08/21/2023

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Coupling Network	EMC PARTNER	CN-EFT1000	KS301188-3	08/22/2022	08/21/2023
Burst Generator	SANKI	EFT-0404S	KUS2009M002-7	03/31/2023	03/30/2024
Coupling and Decoupling Network	SANKI	CDN-4350	KUS2009M002-8	03/31/2023	03/30/2024

Electrostatic Discharge					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
ESD Simulator	EM TEST	DITO 509030	KS301147	02/06/2023	02/05/2024

Radiated Immunity (80MHz-1GHz, 1.4GHz-6GHz)					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
Synthesized Signal Generator	AGILENT	83732B	KS301183	02/03/2023	02/02/2024
Laser probe interface	AR Worldwide	FI7000	KS301193-2	03/21/2023	03/20/2024
E-Field Sensor	AR Worldwide	FL7006 100K-6G	KS301193-1	03/21/2023	03/20/2024
Amplifier Research (80~1000MHz 150w)	AR Worldwide	150W1000M1	KS301139	08/22/2022	08/21/2023
Amplifier Research (1~6GHz 50w)	AR Worldwide	50S1G6M1	KS301231	08/22/2022	08/21/2023
Dual Directional Coupler (1-11G)	AR Worldwide	C1-A47NFNF 35dB	KS301193-5	N.C.R	N.C.R
Dual Directional Coupler (80~1000MHz 400w)	AR Worldwide	DC6180	KS301193-6	N.C.R	N.C.R
RF POWER METER	BOONTON	4232A-01	KS301022	02/03/2023	02/02/2024
POWER SENSOR	BOONTON	51085	H3010235-1	02/03/2023	02/02/2024
POWER SENSOR	BOONTON	51085	H3010235-2	02/03/2023	02/02/2024
Antenna	AR Worldwide	TP1000A	CZ301029	N.C.R	N.C.R
Software	AR	emc ware-v 3.2.0.4	N/A	N/A	N/A

Surge at DC Power Port					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	TRA2006	KS301188-1	08/22/2022	08/21/2023
Coupling and Decoupling Network	EMC PARTNER	CDN-UTP8	KS301188-2	08/22/2022	08/21/2023
Surge Generator	SANKI	LSG-0506S	KUS2009M002-5	03/31/2023	03/30/2024
Coupling and	SANKI	CDN-5350	KUS2009M002	03/31/2023	03/30/2024

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Decoupling Network			-6		

Voltage Dips and Interruptions					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	TRA2006	KS301188-1	08/22/2022	08/21/2023
Switcher	KIKUSUI	SPEC71092	KUS2009M002-2	03/31/2023	03/30/2024
AC Power Supply(Master)	KIKUSUI	PCR24000WE2	KUS2009M002-3	03/16/2023	03/15/2024
AC Power Supply(Slave)	KIKUSUI	PCR24000WE2	KUS2009M002-4	03/16/2023	03/15/2024
Software	KIKUSUI	Quick Immunity Sequencer 2-v 4.0.3.02	N/A	N/A	N/A

General used equipment					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
Digital Pressure Meter	Mengde	DYM3	CZ750023	01/31/2023	01/30/2024
Temperature & Humidity Recorder	Anymetre	TH603	CZ720001-1 CZ720001-2 CZ720001-3 CZ720001-4 CZ720001-5 CZ720001-6 CZ720001-7	10/13/2022	10/12/2023

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6 Immunity Test Results

Performance Criteria Description in EN IEC 61496-1:2020

Criterion A	The EUT shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the EUT is used as intended. If the performance level is not specified by the manufacturer, this may be derived from the product description and documentation and what the user may reasonably expect from the equipment if used as intended.
Criterion B	The EUT shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the EUT is used as intended. The performance level may be replaced by a permissible loss of performance. However, during the test degradation of performance is allowed but no change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the equipment if used as intended.
Criterion C	Temporary loss of function is allowed during the test, provided the function is self-recoverable or can be restored by the operation of the controls.

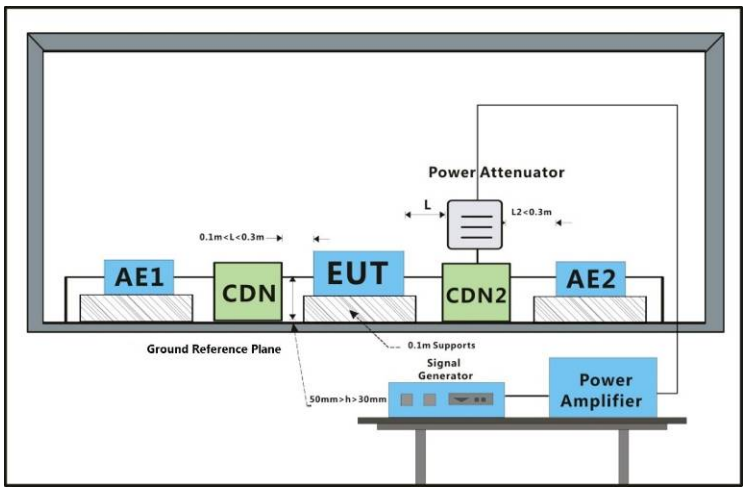
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6.1 Conducted Immunity at DC Power Port (150kHz-80MHz)

Test Requirement: EN IEC 61496-1:2020

Test Method: EN 61000-4-6:2014

6.1.1 Test Setup Diagram



6.1.2 E.U.T. Operation

Operating Environment:

Temperature: 21.5 °C Humidity: 53.6 % RH Atmospheric Pressure: 1010 mbar

6.1.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT Working Normally

6.1.4 Test Condition and Results:

Performance Criterion: A

Frequency Range: 0.15MHz to 80MHz

Modulation: 80%, 1kHz Amplitude Modulation

Step Size 1%

Port	Level (Vrms)	CDN/Clamp	Dwell time	Result / Observations
DC Port	10	CDN	3s	A
A: No degradation in the performance of the EUT was observed				

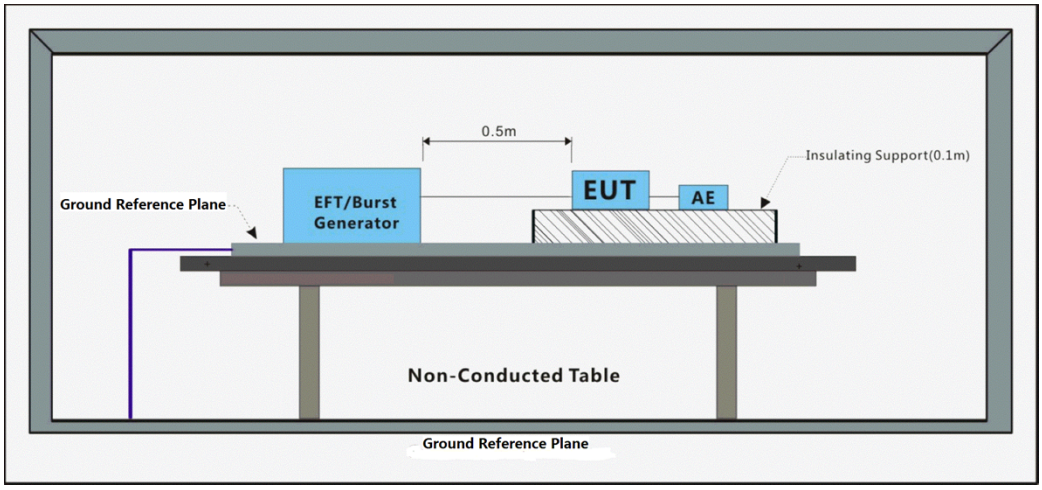
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6.2 Electrical Fast Transients Burst at DC Power Port

Test Requirement: EN IEC 61496-1:2020

Test Method: EN 61000-4-4:2012

6.2.1 Test Setup Diagram



6.2.2 E.U.T. Operation

Operating Environment:

Temperature: 22.4 °C Humidity: 55.6 % RH Atmospheric Pressure: 1010 mbar

6.2.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT Working Normally

6.2.4 Test Condition and Results:

Performance Criterion: B

Repetition Frequency: 5kHz or 100kHz

Burst Period: 300ms

Test Duration: 2 minute per level & polarity

Test Line	Level (kV)	Polarity	CDN/Clamp	Result / Observations
DC Port	1	+	CDN	A
DC Port	1	-	CDN	A
A: No degradation in the performance of the EUT was observed				

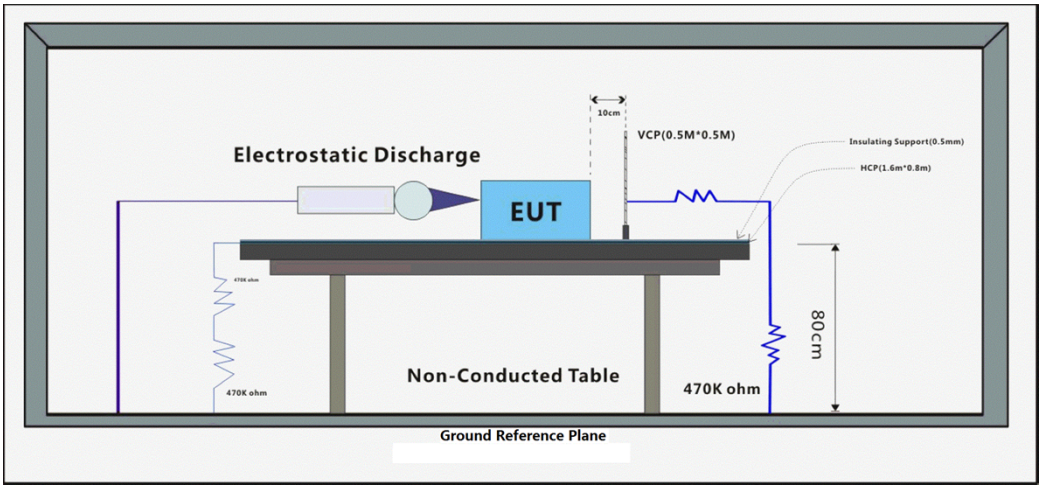
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6.3 Electrostatic Discharge

Test Requirement: EN IEC 61496-1:2020

Test Method: EN 61000-4-2:2009

6.3.1 Test Setup Diagram



6.3.2 E.U.T. Operation

Operating Environment:

Temperature: 25 °C Humidity: 55 % RH Atmospheric Pressure: 1010 mbar

6.3.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT Working Normally

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6.3.4 Test Condition and Results:

Performance Criterion: B
Discharge Impedance: 330 Ω/150pF
Number of Discharge: Minimum 10 times at each test point
Discharge Mode: Single Discharge
Discharge Period: 1 second minimum
Test Point 1: All insulated enclosure & seams.
Test Point 2: All accessible metal parts of the enclosure.
Test Point 3: All sides.

Discharge type	Level (kV)	Polarity	Test Point	Result / Observations
Air Discharge	2,4,8	+	1	A
Air Discharge	2,4,8	-	1	A
Contact Discharge	4	+	2	A
Contact Discharge	4	-	2	A
Horizontal Coupling	4	+	3	A
Horizontal Coupling	4	-	3	A
Vertical Coupling	4	+	3	A
Vertical Coupling	4	-	3	A
A: No degradation in the performance of the EUT was observed				

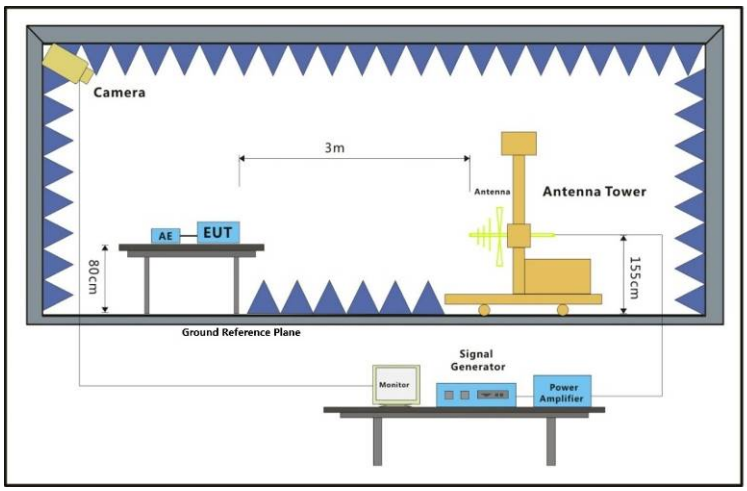
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6.4 Radiated Immunity (80MHz-1GHz, 1.4GHz-6GHz)

Test Requirement: EN IEC 61496-1:2020

Test Method: EN IEC 61000-4-3: 2020

6.4.1 Test Setup Diagram



6.4.2 E.U.T. Operation

Operating Environment:

Temperature: 23.1 °C Humidity: 50.2 % RH Atmospheric Pressure: 1010 mbar

6.4.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT Working Normally

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6.4.4 Test Condition and Results:

Performance Criterion: A

Antenna Polarisation: Vertical and Horizontal

Modulation: 1kHz,80% Amp. Mod,1% increment

Frequency Range: 80MHz to 1GHz, 1.4GHz to 6GHz

Frequency	Level (V/m)	EUT Face	Dwell time	Result / Observations
80MHz-1GHz	10	Front	3s	A
80MHz-1GHz	10	Back	3s	A
80MHz-1GHz	10	Left	3s	A
80MHz-1GHz	10	Right	3s	A
80MHz-1GHz	10	Top	3s	A
80MHz-1GHz	10	Underside	3s	A
1.4GHz-6GHz	3	Front	3s	A
1.4GHz-6GHz	3	Back	3s	A
1.4GHz-6GHz	3	Left	3s	A
1.4GHz-6GHz	3	Right	3s	A
1.4GHz-6GHz	3	Top	3s	A
1.4GHz-6GHz	3	Underside	3s	A
A: No degradation in the performance of the EUT was observed				

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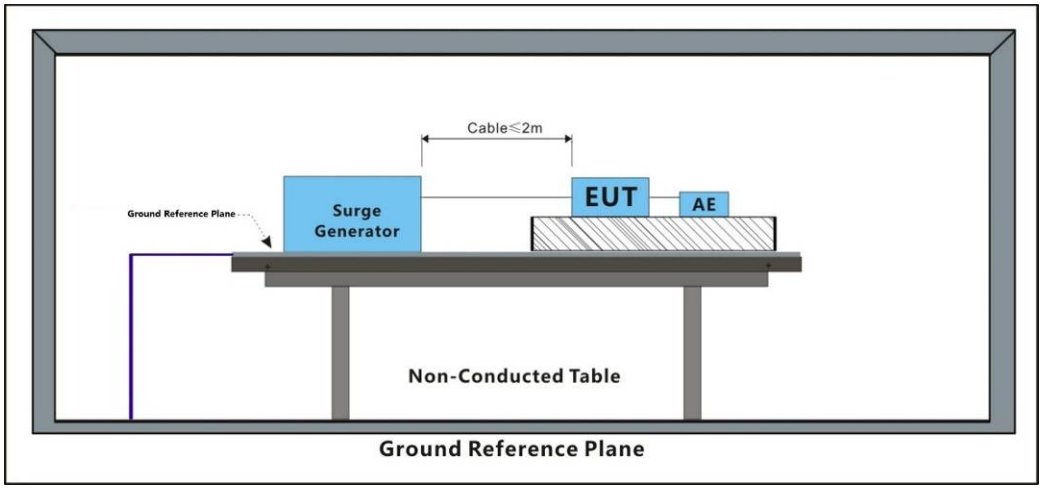
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6.5 Surge at DC Power Port

Test Requirement: EN IEC 61496-1:2020

Test Method: EN 61000-4-5:2014+A1:2017

6.5.1 Test Setup Diagram



6.5.2 E.U.T. Operation

Operating Environment:

Temperature: 21.4 °C Humidity: 55.3 % RH Atmospheric Pressure: 1010 mbar

6.5.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT Working Normally

6.5.4 Test Condition and Results:

Performance Criterion: B

Interval: 60s between each surge

Generator source impedance: 2Ω

CDN coupling impedance(Line-to-ground):10Ω

Test Line	Level (kV)	Polarity	Result / Observations
P-N	0.5	+	A
P-N	0.5	-	A
P-G	1	+	A
P-G	1	-	B
N-G	1	+	B
N-G	1	-	A

A: No degradation in the performance of the EUT was observed

B: During the test,the EUT working abnormally.

After the test,the EUT automatically recovering working normally.

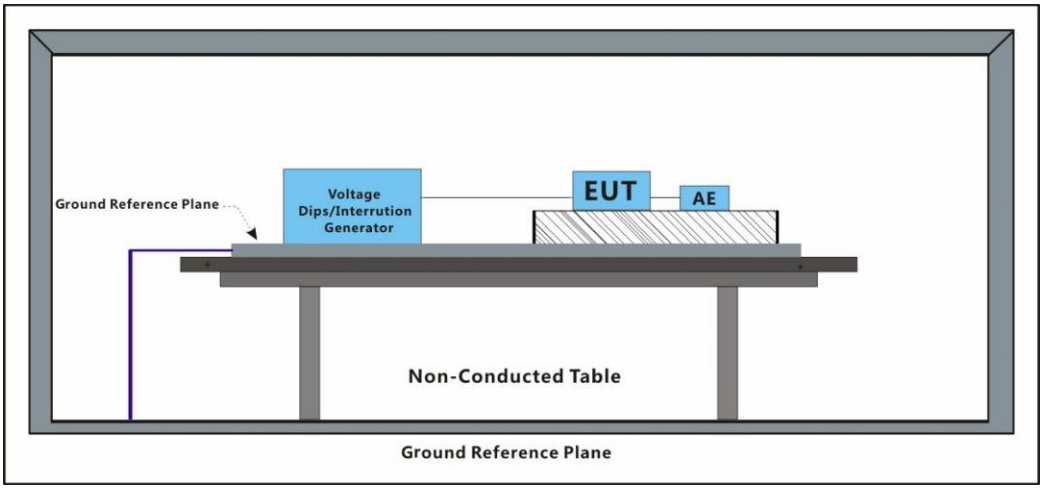
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6.6 Voltage Dips and Interruptions

Test Requirement: EN IEC 61496-1:2020

Test Method: EN 61000-4-29:2000

6.6.1 Test Setup Diagram



6.6.2 E.U.T. Operation

Operating Environment:

Temperature: 21.4 °C Humidity: 55.3 % RH Atmospheric Pressure: 1010 mbar

6.6.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT Working Normally

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6.6.4 Test Condition and Results:

Performance Criterion: 0% of UT (Supply Voltage) for 0.5 Cycle:B;
0% of UT for 1 Cycle:B;
0% of UT for 250 Cycles:C;
70 % of UT for 25 Cycles:C
No. of Dips / Interruptions: 3 per Level
Time between dropout: 10s

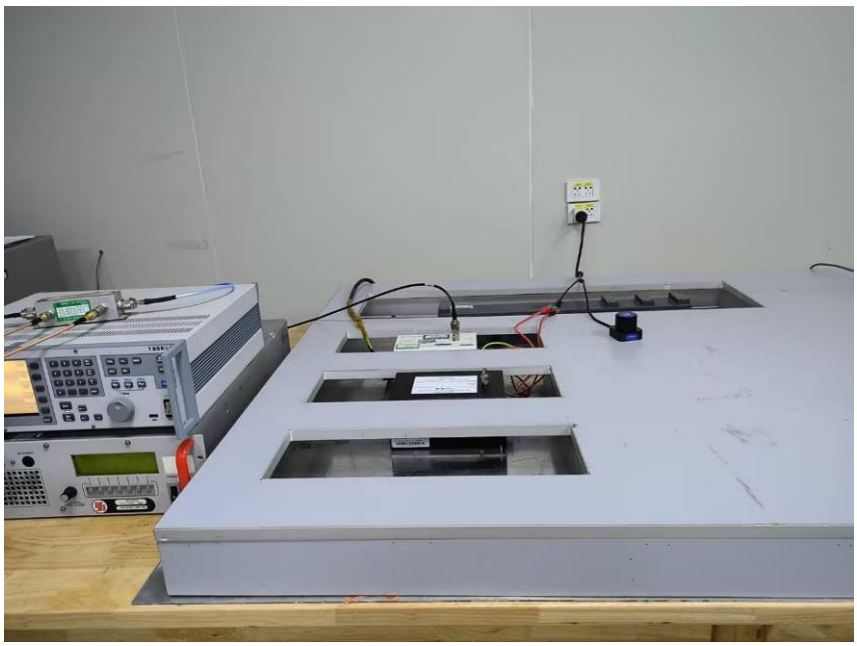
Level % UT	Phase (deg)	Duration	No. of Dips / Interruptions	Result / Observations
0	0°	0.5 Cycle	3	A
0	180°	0.5 Cycle	3	A
0	0°	1 Cycle	3	A
0	180°	1 Cycle	3	A
0	0°	250 Cycles	3	B
0	180°	250 Cycles	3	B
70	0°	25 Cycles	3	A
70	180°	25 Cycles	3	A

A: No degradation in the performance of the EUT was observed
B: During the test,the EUT working abnormally.
After the test,the EUT automatically recovering working normally.

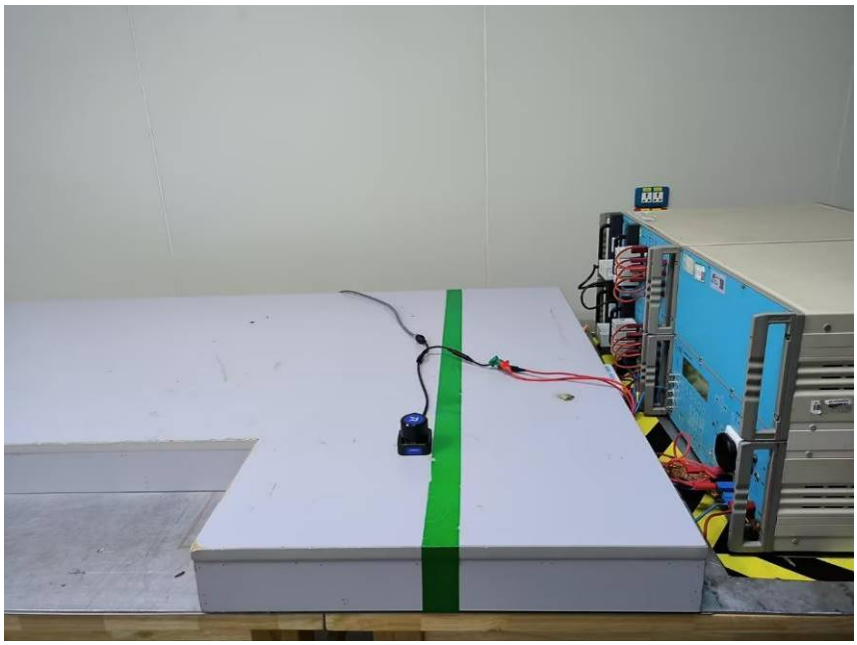
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7 Test Setup Photo

Conducted Immunity at DC Power Port (150kHz-80MHz)

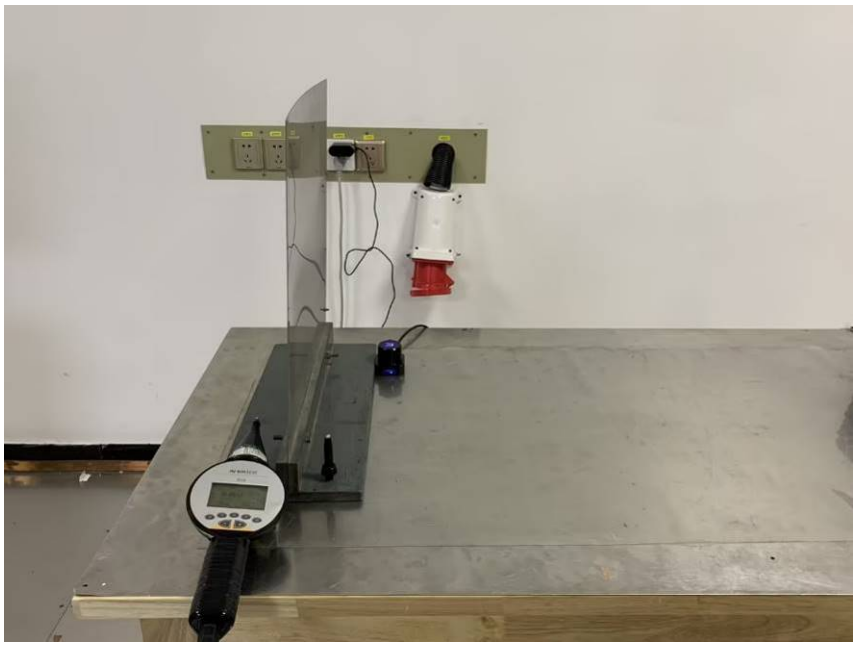


Electrical Fast Transients Burst at DC Power Port

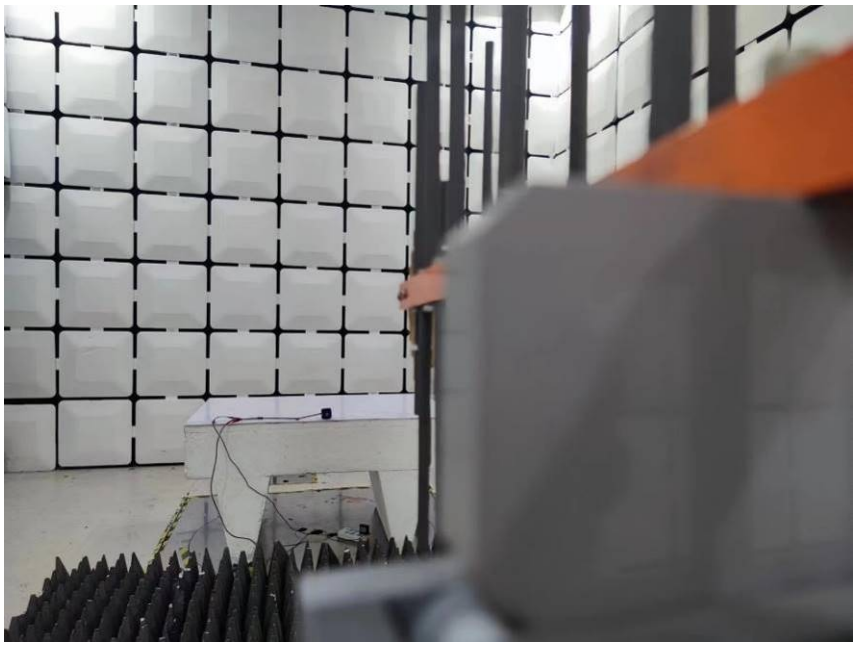


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Electrostatic Discharge

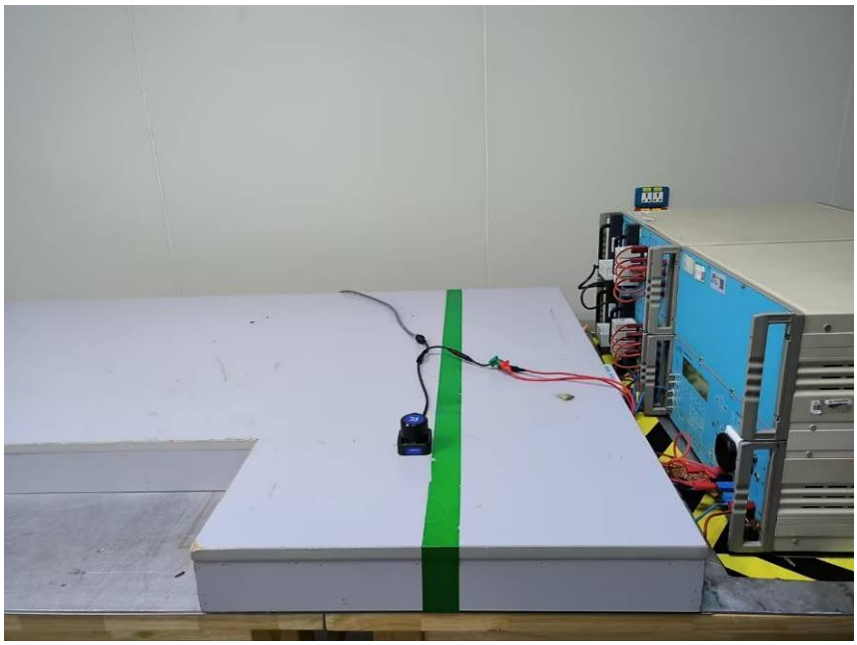


Radiated Immunity (80MHz-1GHz, 1.4GHz-6GHz)



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Surge at DC Power Port



Voltage Dips and Interruptions



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8 EUT Constructional Details (EUT Photos)



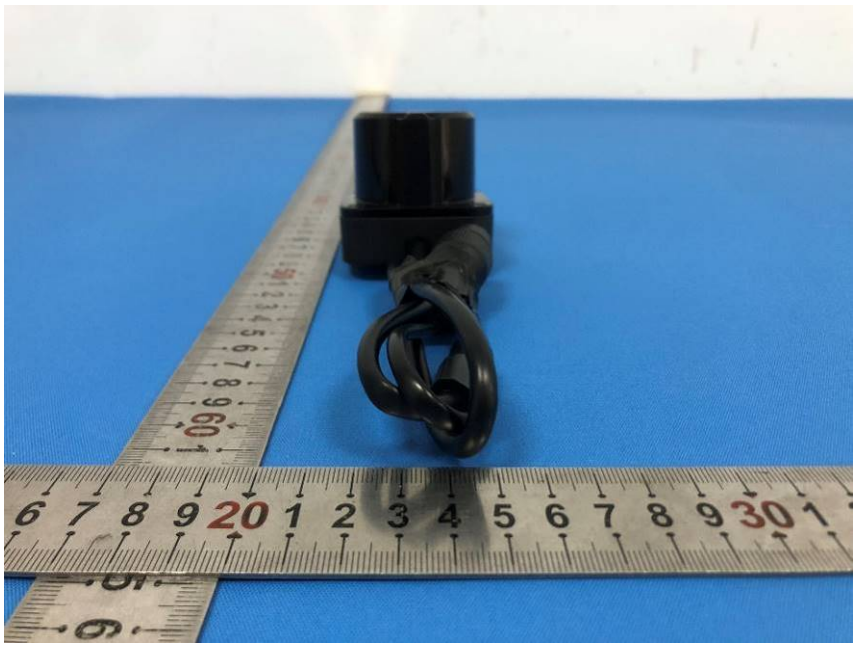


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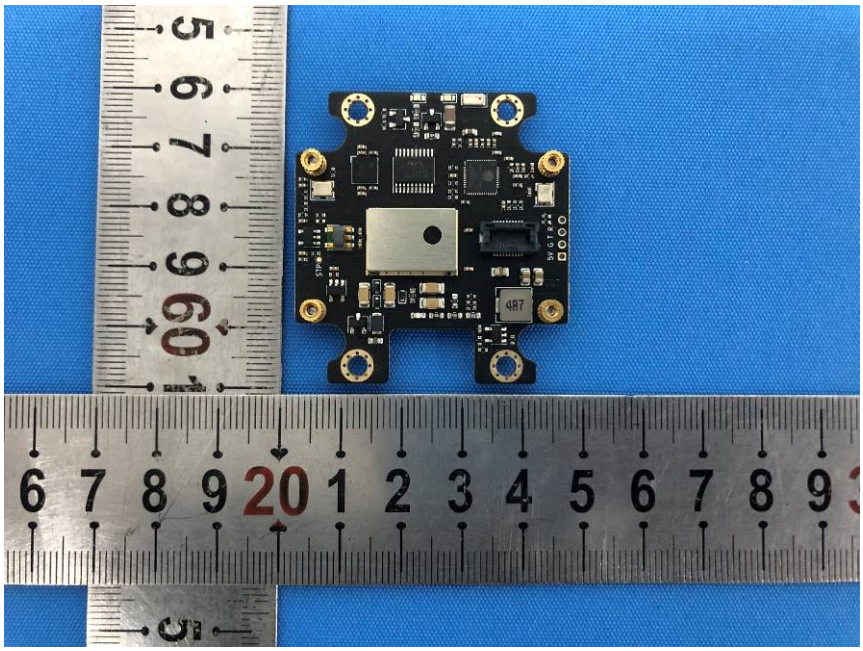




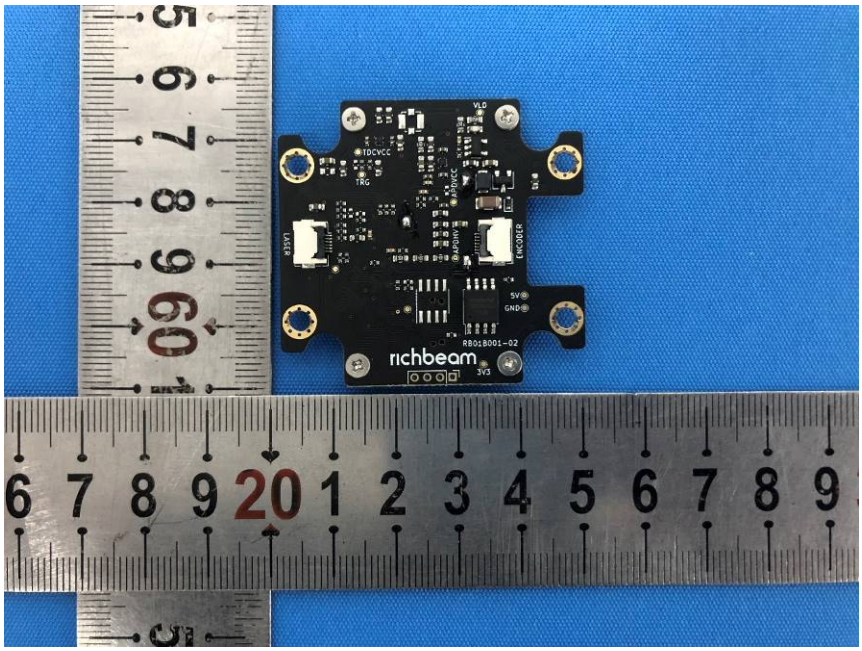
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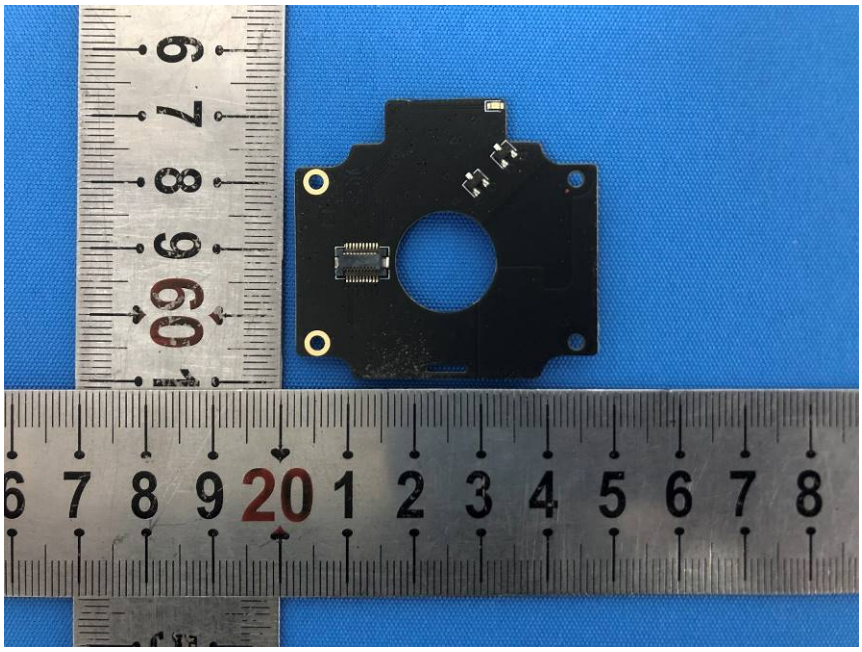
No.10 Weiye Road, Development Zone, Kunshan, Jiangsu, China 215300 t(86-512) 5735 5888 f(86-512) 57370818 www.sgsgroup.com.cn
中国・江苏・昆山开发区伟业路10号 邮编：215300 t(86-512) 57355888 f(86-512) 57370818 sgs.china@sgs.com



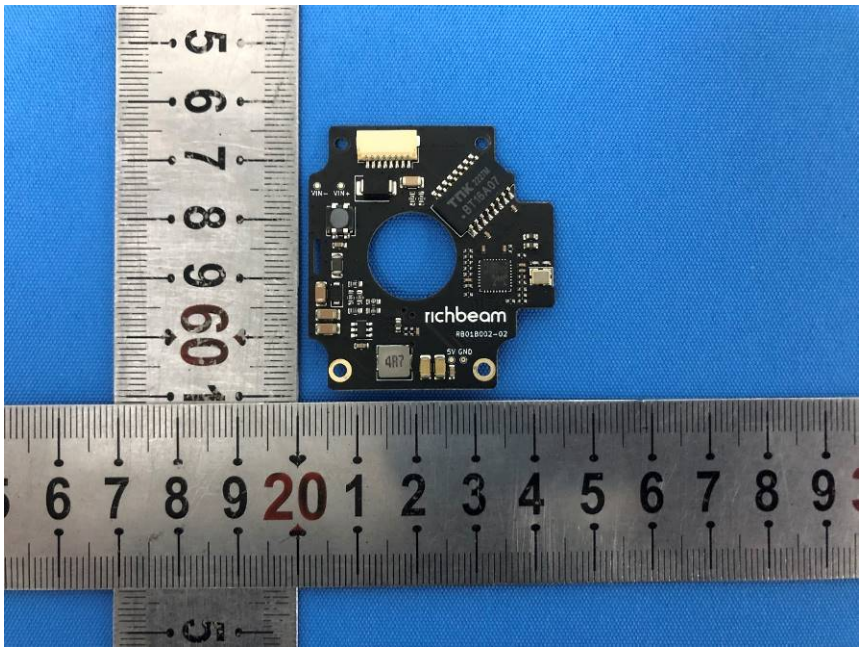


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PR：2D TOF LIDAR
PML：LoraBeam1S
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
(AT 2% Remission)
Response Time ： Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

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PR：2D TOF LIDAR
PML：LoraBeam1H
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
（AT 2% Remission）
Response Time ： Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

PR：2D TOF LIDAR
PML：LoraBeam1
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
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PR：2D TOF LIDAR
PML：KoraBeam1S
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
（AT 2% Remission）
Response Time ： Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

PR：2D TOF LIDAR
PML：KoraBeam1
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
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- End of the Report -

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